

# Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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## Abstract

001 Evaluating AI/ML systems responsibly requires prac-  
002 tices for testing, stress-testing, auditing, comparing,  
003 and interpreting specific capabilities—not only aggre-  
004 gate benchmark scores. This paper proposes such a  
005 practice for spatial reasoning in vision–language mod-  
006 els (VLMs): a diagnostic audit of object-relative ori-  
007 entation, which aggregate accuracy can hide. On Vi-  
008 sual Spatial Reasoning (VSR), moving from a compact  
009 2.2B model to a 7B VLM substantially improves over-  
010 all, depth, horizontal, and containment accuracy, but  
011 orientation changes only modestly; a large 2026 sparse-  
012 MoE model reproduces the pattern (79.5% / 65.7%  
013 overall/orientation), a newer flagship (GPT-5.6 Sol) es-  
014 capes the relative weakness (73.0% orientation), and  
015 a third external model (Gemini Spark) yields only a  
016 degenerate evaluation (chance-level, TRUE-defaulting;  
017 its redo reused the orientation answers, App. A). Ex-  
018 cept for structured prompting, trained conditions on  
019 both controlled models and the sparse-MoE model  
020 remain in a narrow 62–66% orientation band. We  
021 test prompting, general/targeted/hard-negative LoRA,  
022 visual-side adaptation, frozen and object-grounded  
023 probes, and object-centric decomposition; none reliably  
024 closes the gap, and a label-sensitivity analysis (an au-  
025 tomatized audit followed by two blind human re-audits; hu-  
026 mans agree well on the failure taxonomy, but exclusion  
027 masks remain annotator-dependent and automated au-  
028 ditors vary substantially in calibration) shows that the  
029 pattern persists after excluding examples flagged as am-  
030 biguous.

031 Accuracy also fails to capture relational coherence.  
032 LM-side adaptation significantly improves consistency  
033 over zero-shot, while hard-negative training raises  
034 facing/facing-away consistency from 66.0% to 77.7%  
035 with facing accuracy unchanged at 68.9%; the addi-  
036 tional pooled consistency gain over LM-only is not  
037 significant. Cross-dataset evaluation on SITE shows

VSR-trained adaptation does not transfer cleanly:  
neutral overall, significantly harming official spatial-  
relationship reasoning (75.1%→71.6%,  $p = 0.004$ ). A  
pre-specified orientation-vocabulary slice, frozen before  
SITE evaluation, is analyzed as exploratory evidence.  
The results point to an orientation bottleneck that  
is persistent in the controlled model families and the  
sparse-MoE model but not universal (GPT-5.6 Sol es-  
capes; Gemini Spark degenerates), in which orientation  
is not robustly decodable by the tested readouts and  
relational decisions are not always propagated coher-  
ently, while task-specific spatial adaptation can trans-  
fer poorly. Each component of the audit—the diag-  
nostic ladder, the label-sensitivity check, the accuracy–  
coherence separation, and the cross-dataset transfer  
protocol—is specified operationally so that it can be ap-  
plied by others to other capabilities, benchmarks, and  
model families.

## 1. Introduction

This paper contributes an evaluation practice as much  
as a set of model results. We develop a protocol for  
testing, stress-testing, auditing, comparing, and inter-  
preting one capability across vision–language models  
(VLMs), and instantiate it for object-relative orienta-  
tion: facing, facing away, parallel, and perpendicular.  
This capability is relevant to robotics, embodied agents,  
navigation, and scene understanding, but it is easy to  
hide behind an aggregate score: success at identifying  
the relevant objects, inferring the relation, or answering  
complementary formulations does not guarantee suc-  
cess at the others.

Prior benchmarks establish that grounded spatial  
reasoning remains difficult, but they leave open how  
to diagnose a relation-specific failure. VSR intro-  
duced a controlled natural-image probe and reported  
particularly poor object-orientation performance [7];  
What’sUp and GSR-Bench similarly expose failures on

075 basic grounded relations [6, 9]; and broader evaluations  
076 continue to identify orientation as a difficult factor [15].  
077 We therefore ask three narrower questions: does the  
078 VSR orientation weakness persist in modern genera-  
079 tive VLMs and at much larger scale; can prompting,  
080 adaptation, or visual readouts remove it; and do accu-  
081 racy, logical coherence, and cross-dataset transfer tell  
082 the same story?

083 The novelty is diagnostic rather than a claim that  
084 we discovered spatial difficulty. Our contribution is not  
085 any individual diagnostic, but a falsification protocol  
086 that holds the relation family and examples fixed while  
087 testing whether an apparent capability failure survives  
088 scaling and adaptation, label re-audit, representation  
089 readout, logical reformulation, and cross-dataset trans-  
090 fer. Prior work generally reports benchmark accuracy,  
091 studies a single intervention, or proposes a consistency  
092 objective; one controlled audit of the same family dis-  
093 tinguishes a persistent relation-specific bottleneck from  
094 aggregate underperformance, annotation artifacts, or  
095 benchmark-specific specialization.

096 In our VSR experiments, the distinction between  
097 relation families is striking. Moving from SmolVLM2-  
098 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-  
099 creases zero-shot overall accuracy from 74.0% to 80.9%.  
100 Horizontal reasoning rises from 70.2% to 84.8%, depth  
101 from 68.9% to 75.2%, and containment from 83.4%  
102 to 89.3%. Orientation, by contrast, moves only from  
103 62.8% to 63.5%. A large 2026 sparse-MoE model  
104 (MiMo-V2.5, 310B total / 15B active parameters) re-  
105 produces the same relation-specific pattern: 79.5%  
106 overall accuracy yet 65.7% on orientation, again in-  
107 side the 62–66% band. Two further external zero-shot  
108 models bound this result: GPT-5.6 Sol does not repro-  
109 duce the relative weakness—73.0% orientation, above  
110 its own depth (67.1%) and outside the band—while  
111 Gemini Spark’s evaluation degenerates (51.5% over-  
112 all; 98.6% TRUE answers in its first pass, a redo  
113 that reused the 137 orientation verdicts, App. A). The  
114 bottleneck is therefore model-family-dependent rather  
115 than universal, and one function of the protocol is  
116 precisely to detect such differences instead of assum-  
117 ing a failure generalizes. We then evaluate prompting,  
118 language-side adaptation, targeted data construction,  
119 visual-side adaptation, representation probing, and  
120 object-centric decomposition: structured prompting  
121 significantly hurts; general and targeted LoRA improve  
122 aggregate accuracy but do not remove the persistent  
123 orientation weakness; hard negatives improve targeted  
124 relational coherence without a significant orientation-  
125 accuracy gain; and the tested visual-side LoRA config-  
126 urations do not outperform the language-side control.

127 As a sensitivity analysis, removing examples flagged

by the automated audit raises orientation accuracy 128  
(65.7%→78.5% for 7B General LoRA, 66.4%→76.6% 129  
for hard-negative LoRA on the strictest 107-example 130  
subset) but does not eliminate error (2B structured 131  
shows no gain). The exclusion sets are annotator- 132  
dependent: two blind human re-audits agree strongly 133  
with each other on the failure taxonomy (Krippen- 134  
dorff’s  $\alpha=0.81$ ) while their clean/ambiguous flags agree 135  
only moderately with the automated audit ( $\kappa=0.36$  on 136  
the 48 shared persistent-failure cases); details are in the 137  
supplementary. Part of this gain is expected by con- 138  
struction, because the removed examples were drawn 139  
from persistent failures. 140

A second failure mode emerges from logical consis- 141  
tency. The zero-shot 7B model is consistent on only 142  
36.9% of complementary facing/facing-away pairs. LM- 143  
side LoRA substantially improves coherence, and hard- 144  
negative training raises facing consistency to 77.7% 145  
while facing accuracy remains unchanged at 68.9%. 146  
The extra hard-negative gain is not significant in the 147  
pooled strict-family comparison, yet accuracy stays un- 148  
changed while consistency moves: the two metrics cap- 149  
ture different behavior. 150

Finally, we test cross-dataset transfer. The 7B 151  
model is strong on SITE’s official spatial-relationship- 152  
reasoning category (75.1% raw; 59.2% chance-adjusted 153  
accuracy), yet the VSR-trained adapter does not 154  
generalize cleanly: it is statistically neutral overall 155  
(54.2%→53.8%,  $p = 0.66$ ) and significantly degrades 156  
the official category (75.1%→71.6%,  $p = 0.004$ ). A 157  
pre-specified, non-official orientation-vocabulary slice 158  
is much weaker in the aggregate (48.3% raw; 24.0% 159  
chance-adjusted) and shows no positive transfer, but— 160  
as we detail below—its composition limits what it can 161  
establish, and we therefore do not treat SITE as inde- 162  
pendent confirmation of the VSR orientation construct. 163

Our contributions are four reusable components of 164  
an evaluation protocol: 165

- A fine-grained diagnostic ladder. Across ten VSR 166  
conditions, two differently sized models, and three 167  
external zero-shot models, several spatial fami- 168  
lies improve substantially while object-relative ori- 169  
entation remains unusually resistant in the con- 170  
trolled families; one newer flagship escapes the rel- 171  
ative weakness and one external model’s evaluation 172  
degenerates under the protocol, bounding rather 173  
than weakening the claim. The ladder—scaling, 174  
prompting, language-side adaptation, targeted data 175  
construction, visual-side adaptation, representation 176  
probing, and explicit decomposition—is defined op- 177  
erationally so it transfers to other benchmarks and 178  
relation families. 179
- A robustness and sensitivity audit. A label- 180

|     |                                                            |     |
|-----|------------------------------------------------------------|-----|
| 181 | sensitivity analysis (automated labels with two inde-      | 232 |
| 182 | pendent human re-audits), frozen/object-grounded           | 233 |
| 183 | probes, visual-side adaptation, and object-centric         | 234 |
| 184 | decomposition constrain purely annotation-based or         | 235 |
| 185 | single-component explanations. The audit practice          | 236 |
| 186 | (flagged-example exclusion sets, decodability checks       | 237 |
| 187 | against majority baselines) applies to any bench-          | 238 |
| 188 | mark whose labels or examples may be ambiguous.            | 239 |
| 189 | • An accuracy-coherence separation protocol.               | 240 |
| 190 | Complementary-relation tests on VSR's strict               | 241 |
| 191 | relation pairs show that, for the tested conditions,       | 242 |
| 192 | relational consistency can change substantially            | 243 |
| 193 | while task accuracy remains fixed. The protocol            | 244 |
| 194 | measures consistency against verdict-pattern base          | 245 |
| 195 | rates, so it can be applied wherever complementary         | 246 |
| 196 | formulations of the same relation exist.                   | 247 |
| 197 | • A cross-dataset transfer protocol. SITE shows            | 248 |
| 198 | strong official spatial-relationship performance,          | 249 |
| 199 | while VSR-trained LoRA transfers poorly and sig-           | 250 |
| 200 | nificantly harms the official SITE category; the           | 251 |
| 201 | orientation-vocabulary slice is analyzed as ex-            | 252 |
| 202 | ploratory evidence. The protocol (pre-specified            | 253 |
| 203 | subsets, chance-adjusted accuracy, paired transfer         | 254 |
| 204 | tests) is model- and benchmark-agnostic and is re-         | 255 |
| 205 | ported with negative as well as positive transfer.         |     |
| 206 | The protocol is deliberately modular: each compo-          |     |
| 207 | nent can be applied independently to other capabilities,   |     |
| 208 | benchmarks, and model families. The supplementary          |     |
| 209 | specifies the prompts, subsets, decision rules, and re-    |     |
| 210 | leased artifacts needed to reproduce the audit.            |     |
| 211 | 2. Related Work                                            |     |
| 212 | Benchmarks for grounded spatial reasoning. VSR in-         |     |
| 213 | troduced a controlled natural-image benchmark with         |     |
| 214 | 66 spatial relations and identified facing and parallel as |     |
| 215 | difficult relations for the VLMs studied at the time [7].  |     |
| 216 | What'sUp controls object identity while varying spa-       |     |
| 217 | tial configuration and reports a substantial human-        |     |
| 218 | model gap [6]; GSR-Bench broadens grounded evalu-          |     |
| 219 | ation across models and scales [9]. SITE extends cov-      |     |
| 220 | erage to single-image, multi-image, and video inputs       |     |
| 221 | and organizes tasks by dimensions such as orientation      |     |
| 222 | and intrinsic/extrinsic relations [15]. SpatialScore and   |     |
| 223 | Mind the Gap further show that spatial competence          |     |
| 224 | remains fragmented [10, 16]. We use VSR not to re-         |     |
| 225 | establish that spatial reasoning is hard, but to turn its  |     |
| 226 | orientation signal into a controlled diagnostic target     |     |
| 227 | and use SITE to test transfer.                             |     |
| 228 | Diagnosing and improving spatial failures. Several         |     |
| 229 | lines of work move beyond aggregate accuracy.              |     |
| 230 | AdaptVis relates successful VSR reasoning to attention     |     |
| 231 | on relevant object locations [2], while CREG shows that    |     |
|     | attribution evidence in Qwen2-VL can reflect image lay-    |     |
|     | out rather than the queried relation [11]. When More Is    |     |
|     | Less finds that spatial cues, commonsense information,     |     |
|     | and reasoning instructions can help or hurt depend-        |     |
|     | ing on the model and setup [1]. Other work probes          |     |
|     | visual-concept decodability [13], separates perception     |     |
|     | from reasoning in spatial planning [17], or supplies ge-   |     |
|     | ometric inputs for 3D spatial understanding [3]. These     |     |
|     | motivate our combination of frozen/readout diagnos-        |     |
|     | tics with adaptation and object-centric decomposition,     |     |
|     | but none audits one relation family across accuracy,       |     |
|     | coherence, label sensitivity, and transfer.                |     |
|     | Taken together, prior work establishes the problem         |     |
|     | and offers several useful diagnostic lenses, but leaves    |     |
|     | a methodological gap: benchmark accuracy, represen-        |     |
|     | tation access, logical consistency, annotation sensitiv-   |     |
|     | ity, and transfer are usually reported separately. We      |     |
|     | close that gap with a single controlled diagnostic lad-    |     |
|     | der, with analysis choices and external-evaluation sub-    |     |
|     | sets frozen where specified, applied to the same rela-     |     |
|     | tion family. Our claim is consequently not to discover     |     |
|     | spatial difficulty, but to make a persistent orientation   |     |
|     | weakness measurable, falsifiable, and reusable as an       |     |
|     | evaluation protocol.                                       |     |
|     | Adaptation, specialization, and transfer. Parameter-       |     |
|     | efficient adaptation such as LoRA [5] can improve a        |     |
|     | benchmark without improving the underlying capabil-        |     |
|     | ity. We therefore compare general, targeted, hard-         |     |
|     | negative, projector, and vision-plus-projector updates,    |     |
|     | then evaluate the strongest language-side adapter on       |     |
|     | an untouched benchmark. This design distinguishes          |     |
|     | within-benchmark specialization from transfer: VSR         |     |
|     | adaptation is neutral overall on SITE, does not im-        |     |
|     | prove the orientation subset, and significantly degrades   |     |
|     | SITE's official spatial-relationship category. Our em-     |     |
|     | phasis on negative transfer complements prior bench-       |     |
|     | mark work by treating transfer failure as an evaluation    |     |
|     | result rather than an omitted outcome.                     |     |
|     | Representation probing and logical consistency.            |     |
|     | Probes can test whether a signal is accessible from        |     |
|     | frozen representations, but probe capacity can itself      |     |
|     | learn the task, requiring control baselines and cautious   |     |
|     | interpretation [4]. Consistency-oriented objectives        |     |
|     | such as SAGE use geometric and linguistic dualities        |     |
|     | to train spatial reasoning [8]. We instead use probing     |     |
|     | and complementary-pair consistency as independent          |     |
|     | diagnostics: the former asks what the tested readouts      |     |
|     | can decode, while the latter asks whether answers          |     |
|     | to logically linked statements agree, so accuracy and      |     |
|     | coherence can diverge as diagnostic axes.                  |     |

## 282 3. Experimental Setup

### 283 3.1. Benchmarks

284 VSR. We use VSR [7] as the primary diagnostic  
285 benchmark. The random-split test set contains 2,195  
286 examples. The benchmark defines 64 relation labels;  
287 61 appear in the evaluated test split. The orientation  
288 family contains facing, facing away from, parallel to,  
289 and perpendicular to, totaling 137 test examples. All  
290 main VSR results use the same held-out test set and  
291 deterministic True/False parsing.

292 SITE. We use SITE [15] for cross-dataset transfer  
293 evaluation only and do not train on SITE. The image  
294 evaluation contains 2,591 examples: 1,368 single-image  
295 and 1,223 multi-image examples. Our primary SITE  
296 subset is the official spatial-relationship-reasoning cate-  
297 gory ( $n = 993$ ). We additionally report a transparently  
298 defined, non-official keyword-derived orientation subset  
299 ( $n = 2,024$ ), analyzed as exploratory evidence rather  
300 than as direct replication of the VSR construct. SITE  
301 results use the benchmark’s native multiple-choice for-  
302 mat and chance-adjusted accuracy (CAA) in addition  
303 to raw accuracy.

### 304 3.2. Models and Conditions

305 Our compact model is SmolVLM2-2.2B-Instruct, and  
306 our larger controlled-study model is Qwen2-VL-7B-  
307 Instruct [14]. We retain Qwen2-VL-7B as the con-  
308 trolled anchor because it supports the complete local  
309 ladder—the same base model across prompting, LoRA,  
310 visual-side adaptation, probing, and decomposition—  
311 rather than mixing intervention effects with an uncon-  
312 trolled architecture change. We evaluate ten canon-  
313 ical VSR conditions: 2B zero-shot, 2B structured  
314 prompting, 2B General LoRA, 2B Targeted LoRA,  
315 7B zero-shot, 7B General LoRA, 7B Targeted LoRA,  
316 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B  
317 Vision+Projector LoRA. LoRA [5] is used for all  
318 parameter-efficient adaptation conditions. We addi-  
319 tionally evaluate a large zero-shot sparse-MoE model,  
320 MiMo-V2.5 [12] (310B total / 15B activated param-  
321 eters, dedicated 729M-parameter vision encoder, 2026),  
322 served through its public API with the identical  
323 frozen prompt, thinking disabled, and greedy decoding  
324 (App. A). Its role is external validation of the diagnosed  
325 pattern; it is not fine-tuned. Two additional external  
326 zero-shot models used the same frozen prompt: GPT-  
327 5.6 Sol (OpenAI, interactive, 2026-08-12) and Gem-  
328 ini Spark (Google, run\_gemini\_spark\_vsr2195\_001).  
329 Gemini’s first pass defaulted to TRUE on 98.6% of  
330 items; its redo reused the 137 orientation answers  
331 (id-level verified), so its Table 1 row is a degenerate-

evaluation finding, not a capability estimate (App. A).

General LoRA uses a proportional sample across  
relation families; Targeted LoRA concentrates train-  
ing on weak orientation/depth/horizontal families with  
rehearsal of the rest; Hard-Neg LoRA adds au-  
dited complementary orientation examples, primarily  
facing/facing-away inversions; vision-side conditions  
adapt the multimodal projector or upper vision blocks  
with it.

### 331 3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial fam-  
ily. For SITE we report raw accuracy and CAA. Paired  
condition comparisons use McNemar tests when ap-  
plicable. Clean-label orientation analysis reports the  
full orientation set and progressively stricter audited  
subsets. Logical consistency is evaluated by construct-  
ing complementary statements for strict relation pairs  
(left/right, front/behind, facing/facing-away) and test-  
ing whether the model assigns opposite verdicts as re-  
quired. Parallel/perpendicular is treated separately as  
a soft complement because both relations can be false  
for oblique objects.

## 354 4. Main Results

### 355 4.1. Orientation Remains Weak Across Two Differ- 356 ently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B  
model to the 7B model lifts zero-shot overall accuracy  
by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and  
containment by 5.9 pp, but orientation by only 0.7 pp.  
General 7B LoRA further raises overall accuracy to  
84.7%, yet orientation reaches only 65.7%. Across  
trained conditions other than structured prompting,  
orientation remains in a narrow 62–66% band. The  
large sparse-MoE zero-shot model (MiMo-V2.5, evalu-  
ated identically, App. A) falls in the same band: 79.5%  
overall and 65.7% orientation, with the orientation fam-  
ily again its weakest relation family (per-family table  
in the supplementary). The relation-specific pattern  
is therefore not an artifact of the 2B/7B scale range.  
Two further external zero-shot models bound the pat-  
tern: GPT-5.6 Sol does not reproduce the relative  
weakness (73.0% orientation vs. its own 67.1% depth;  
point estimate outside the band, 95% CI [65.0, 79.7]  
overlapping it), while Gemini Spark’s evaluation degen-  
erates (51.5% overall; App. A). The MiMo-V2.5 row  
is also robust to output-format parsing: orientation  
is unchanged parseable-only (65.7%), and parseable-  
only overall is 82.1% vs. 79.5% counting the 69 non-  
conforming outputs wrong (App. G).

To quantify the family-relative claim directly,



Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

| Condition                           | Overall | Orientation | Depth | Horizontal | Containment | Topology/contact |
|-------------------------------------|---------|-------------|-------|------------|-------------|------------------|
| 2B zero-shot                        | 74.0    | 62.8        | 68.9  | 70.2       | 83.4        | 80.5             |
| 2B structured                       | 68.3    | 53.3        | 64.9  | 66.7       | 78.7        | 71.7             |
| 2B General LoRA                     | 76.6    | 62.0        | 71.1  | 74.3       | 87.6        | 81.0             |
| 2B Targeted LoRA                    | 76.5    | 64.2        | 70.8  | 75.1       | 87.6        | 81.4             |
| 7B zero-shot                        | 80.9    | 63.5        | 75.2  | 84.8       | 89.3        | 80.3             |
| 7B General LoRA                     | 84.7    | 65.7        | 82.3  | 87.4       | 92.9        | 84.4             |
| 7B Targeted LoRA                    | 83.9    | 64.2        | 81.7  | 87.1       | 91.7        | 85.3             |
| 7B Hard-Neg LoRA                    | 84.3    | 66.4        | 80.7  | 87.1       | 89.9        | 84.6             |
| 7B Projector LoRA                   | 82.9    | 64.2        | 78.6  | 87.1       | 89.3        | 85.3             |
| 7B Vision+Projector LoRA            | 83.1    | 64.2        | 78.0  | 87.1       | 89.9        | 84.4             |
| MiMo-V2.5 zero-shot                 | 79.5    | 65.7        | 74.8  | 81.9       | 87.1        | 84.6             |
| GPT-5.6 Sol zero-shot               | 75.4    | 73.0        | 67.1  | 76.9       | 82.8        | 77.8             |
| Gemini Spark zero-shot <sup>†</sup> | 51.5    | 57.7        | 55.0  | 53.2       | 50.3        | 47.5             |

<sup>†</sup>Gemini Spark’s evaluation is degenerate (98.6% TRUE in pass 1; redo reused the orientation answers, id-level verified); the row is a degenerate-evaluation finding, not a capability estimate (App. A).

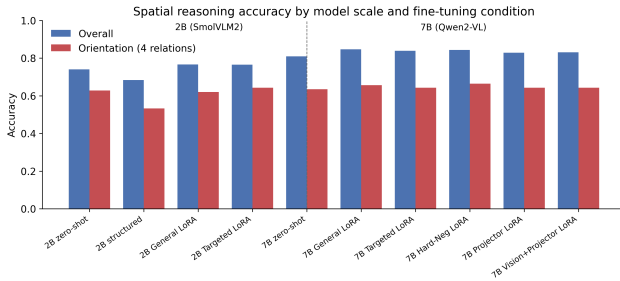


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

a paired bootstrap over the 2,195 shared examples (10,000 resamples) confirms the horizontal-vs-orientation scaling contrast is robust (+13.9 pp; 95% CI [+2.9, +24.8]; positive in 99.3% of resamples), with depth (+5.5 pp) and containment (+5.1 pp) directionally positive but not individually significant (App. D).

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ( $p \approx 2.2 \times 10^{-7}$ ). Consistent with recent VSR analyses where additional cues or reasoning instructions help or hurt depending on setup [1], this is specific to the tested 2B condition and prompt.

## 4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; frozen v1) flagged ambiguous or questionable orientation examples; we repeat evaluation on progressively stricter sub-

sets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these are not matched comparisons against other relation families. Two blind human re-audits under the same taxonomy (binary flag on all 137, eight-class taxonomy on the 48 persistent failures) yield three distinct analyses, reported separately.

Taxonomy-derived strict masks. Each rater’s strict mask raises orientation accuracy without eliminating error—automated 78.5% ( $n=107$ ), human A 77.6% ( $n=107$ ), human B 78.8% ( $n=104$ )—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag ( $\kappa=0.645$ , 82.5% agreement) but only moderately with the automated audit ( $\kappa=0.36$  on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels ( $\alpha=0.14$ ). Under the human binary masks the clean-subset gains largely vanish (best 66.7% and 66.2% vs. the 66.4% full-set best), tying the gains to the taxonomy-derived masks rather than to the binary flag. Audit reliability is itself model-dependent: two further LLM auditors (GPT-5.6 Sol and Gemini Spark) agree with the humans only weakly ( $\kappa=0.34$  and  $\kappa=0.03$  on the 137-example flag; Gemini flags 126/137 as clean), so LLM-generated ambiguity labels are sensitivity tools, not authoritative ground truth (App. F).

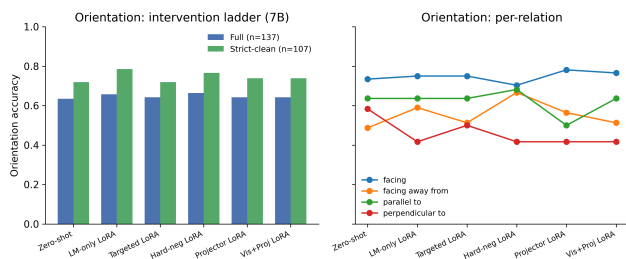


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

Consensus analysis. Under the consensus union mask ( $n=62$ ), the trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and the 2B structured no-gain pattern does not survive this mask (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 projector LoRA and both human tables in the supplementary).

## 5. Why Does Orientation Remain Difficult?

### 5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ( $n=34$ ) and the confidence interval overlaps the majority baseline.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Probing decodability of visual concepts has precedent in driving-oriented VLMs [13] and attribution analyses of this model family [11]; ours tar-

gets VSR’s orientation family with pooled, patch-vote, and object-grounded readouts over the model’s exact conditioning features, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

### 5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted  $p=0.0043/0.0122$ ; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not close the orientation gap.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts match the end-to-end control’s correctness on only 55.1–58.3% of the 127 boxed orientation statements (per-relation agreement 25.0–69.4%, weakest for parallel; App. D); in the committed analysis (10 no-box statements scored wrong) the verdicts diverge from the control on 61–65 of the 137 statements, with the control correct and the pipeline wrong in 44–47 of them. These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control’s verdicts, so we report the tested decomposition as an agreement diagnostic that fails to reproduce the generative model’s behavior, not as a direct accuracy comparison (ground-truth scoring via the released script, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

### 5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair consistency has prior liter-

ature, including consistency-oriented training [8]; our diagnostic below uses strict pairs with verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE zero-shot model does not repair coherence either: it reaches 42.7% facing-pair consistency under the same definition (vs. 36.9% for the 7B zero-shot model; 44.0% on the 100 pairs with parseable verdicts, three excluded for non-conforming outputs), with the same False-defaulting verdict pattern as the smaller models (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The two external models diverge: GPT-5.6 Sol reaches 83.3% complementary facing verdicts (no both-True,  $n=102$ ; 54.5% expected under independence), far above the controlled families, while Gemini Spark’s 60.2% with a zero flip-True rate and 100% complementary front/behind pairs are treated as degenerate, not capability estimates (App. A). The observed verdict patterns show a False-defaulting bias: both-False verdicts on 63.1% of pairs, no both-True verdicts (0/103), and expected consistency under verdict independence at 30.2%, below the observed 36.9% (verdict table in the supplementary)—a refusal to affirm either proposition rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot,  $p < 0.0001$ ); hard-negative training raises facing consistency to 77.7% with facing accuracy identical at 68.9% for both, although the additional pooled consistency effect over LM-only is not significant ( $p = 0.29$ ). The key result is separability: identical facing accuracy coexists with substantially different facing consistency; we do not claim a general coherence algorithm. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

## 6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained

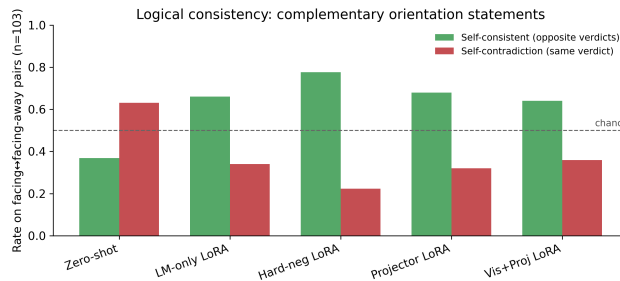


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%,  $p = 0.66$ ) and significantly degrades the official category (75.1%→71.6%,  $p = 0.004$ ; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category) and shows no positive transfer under the adapter (48.3%→48.7%,  $p = 0.70$ ). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16],  $p = 0.29$ ), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%,  $n=488/505$ ). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular;  $n=177$ ) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

## 7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weak-

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

| Subset                | Zero-shot   | VSR-LoRA    | $\Delta$ |
|-----------------------|-------------|-------------|----------|
| All images            | 54.2 / 31.1 | 53.8 / 30.5 | -0.4     |
| Official spatial-rel. | 75.1 / 59.2 | 71.6 / 53.4 | -3.5     |
| Orient. heuristic     | 48.3 / 24.0 | 48.7 / 24.5 | +0.4     |

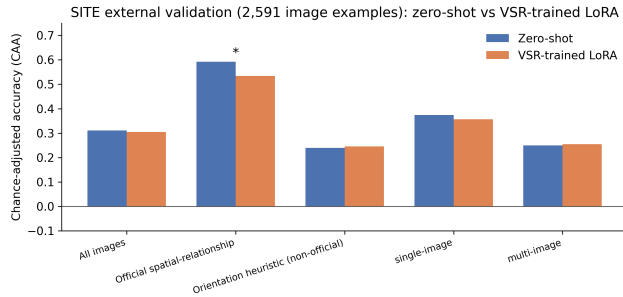


Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

ness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation [7] while extending it to modern generative VLMs and a broader intervention analysis. The weakness is also model-family-dependent: the controlled families and the sparse-MoE model persist in it, GPT-5.6 Sol does not reproduce it, and Gemini Spark’s evaluation degenerates under the protocol; a diagnostic that only found the failure would be uninformative, so this distinguishability is what makes the protocol falsifiable.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and signif-

icantly negative on the official subset. Hard negatives improve a targeted coherence behavior with relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated with general spatial-reasoning improvement.

## 8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has  $n=12$ ; perpendicular results are descriptive only. The Wilson 95% confidence intervals on the family-level point estimates are correspondingly wide—62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA ( $n=137$  each)—so the trained conditions’ 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%,  $n=34$ ; 95% CI [57, 85] vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%,  $n=12$ ; [19, 68] vs. 50%), and the 7B zero-shot facing-away subset (48.7%,  $n=39$ ; [34, 64]). Paired McNemar tests on orientation rest on small discordant-pair counts: the non-significant projector and vision+projector comparisons ( $p=0.86$ ,  $p=0.85$ ) indicate no detectable effect, not equivalence; conclusions rely on the full-set paired tests and the Holm-adjusted battery (supplementary).

The clean-label audit’s exclusion sets originate from an automated LLM audit: two independent human re-audits reached only moderate per-item agreement with it ( $\kappa=0.36$  on the clean/ambiguous flag; Krippendorff’s  $\alpha=0.14$  on the eight-class taxonomy) while agreeing strongly with each other on the failure taxonomy ( $\alpha=0.81$ ); both reproduced the qualitative conclusions under their own taxonomy-derived strict masks, so the sets are annotator-dependent (tables versioned: frozen v1 plus both human re-audits). The SITE orientation slice is keyword-derived; a composition-confound analysis shows its aggregate score is largely explained by task composition, so SITE is not independent confirmation of the VSR bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full retraining; the three model families form a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: paired McNemar tests quantify disagreement on the fixed test ex-



amples, not training-run variability (a multi-seed extension is prepared in scripts/run\_seed\_variance.py). The MiMo-V2.5 evaluation is served through a public API: 3.1% of outputs did not conform to the one-word format and are counted wrong (never guessed), and 294 contained chain-of-thought text alongside the verdict; both are disclosed in the committed predictions (App. A). The MiMo-V2.5 row was evaluated on 2026-08-12 through the OpenCode Go gateway; API-served models can be updated without notice, so this row’s reproducibility depends on the serving snapshot (evaluation date, model id, and per-request token accounting are recorded in results/mimo/). The two additional external rows (App. A) used the identical frozen prompt via third-party interfaces on 2026-08-12; GPT-5.6 Sol’s interactive run carries no deterministic-decoding guarantee, and Gemini Spark’s evaluation is degenerate (98.6% TRUE in pass 1; redo reused 1,315/2,195 answers including all 137 orientation verdicts, id-level verified), so its row and coherence numbers are a documented degenerate-evaluation outcome, not a capability estimate. The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results; lack of improvement holds for the tested configurations, not as impossibility claims.

## 9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the studied model families: larger models and standard adaptation improve several families while leaving orientation resistant, a large sparse-MoE model reproduces the pattern, a newer flagship escapes it, and one external model’s evaluation degenerates under the protocol—evidence the weakness is model-family-dependent, not universal. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations; complementary-relation tests show accuracy and coherence are distinct axes that both remain weak in the large model. SITE transfer shows VSR-specific adaptation can negatively transfer. Evaluation should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy.

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